

GRADUATE SEMINAR READING RESPONSE | WEEK 7

High-Latency Fault Tolerance & Consensus Protocols

CS 8803: Advanced Distributed Infrastructure & Network Resilience

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Track: Distributed Systems

1 Assigned Readings & Metadata

Reading Portfolio

Ref	Title & Authors	Venue / Year	Primary Focus
[P1]	Speculative State Replication in Partial Asynchrony <i>Marcus Thorne, Elena Rostova, and Julian Sterling</i>	ACM TOCS (2024)	Consensus Latency
[P2]	Dynamic Quorum Reconfiguration under Byzantine Churn <i>Sarah Jenkins and David K. Chen</i>	OSDI '25	Adaptive Membership
[P3]	Practical Bounded Memory Footprints for Geo-Raft <i>Nexa Research Group (Tech Report #402)</i>	Tech Report (2026)	Storage Overhead

🔑 Executive Summary of Week 7 Scope: This week's readings examine the tension between network latency and safety guarantees in fault-tolerant replicated state machines. **[P1]** introduces optimistic fast-path execution reducing round-trips from $3 \rightarrow 1$ under zero-contention scenarios. **[P2]** relaxes static node set assumptions by formalizing epoch-based dynamic reconfiguration under active adversarial churn. **[P3]** addresses log compaction memory explosion in multi-region deployments managed by Synapse Infrastructure Engine.

2 Core Synthesis & Comparative Analysis

2.1 Architectural Paradigms: Optimistic vs. Pessimistic Consensus

The central thesis connecting **[P1]** and **[P2]** revolves around the tradeoff between fast-path speculation and worst-case protocol fallback. Traditional Multi-Paxos and Raft enforce structural ordering prior to state application. Thorne et al. (**[P1]**) break this barrier by allowing replicas to speculatively execute incoming request batches provided a supermajority ($Q_{\text{fast}} = \lceil \frac{3f+1}{2} \rceil$) of nodes agree on the sequence vector.

Mathematically, while classical PBFT requires $3f + 1$ total nodes to tolerate f Byzantine faults with 2-phase commit (2 message round-trips), **[P1]** achieves single round-trip consensus whenever the network latency ratio satisfies:

$$\Delta_{\text{actual}} \leq \frac{1}{2} \delta_{\text{estimated}} + \gamma_{\text{jitter}} \quad (1)$$

where γ_{jitter} represents the variance in inter-replica packet transit time across WAN links.